OpenTelemetry Foundations & OTCA Prep - Hands-on

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Primary Links

- OpenTelemetry Documentation
- OpenTelemetry Certified Associate (OTCA) — Linux Foundation
- hello-telemetry — GitHub Repo
- OpenTelemetry Demo Documentation
- OpenTelemetry Demo — GitHub Repo
- OpenTelemetry Java Documentation
- OpenTelemetry Python Documentation
- OpenTelemetry Java Examples — GitHub Repo
- OpenTelemetry Concepts
- Semantic Conventions — GitHub Repo

Links by Section

Section 2: Observability

- OpenTelemetry Vendors

Section 3: Learning Environment

- Visual Studio Code
- Docker Desktop
- Microsoft Build of Open JDK
- Apache Maven
- Python
- hello-telemetry — GitHub Repo
  - Web page: http://localhost:8085/MyWebApp
  - Jaeger: http://localhost:16686/search
- OpenTelemetry Demo — GitHub Repo
- OpenTelemetry Demo — Docker setup guide
- OpenTelemetry Demo links:
  - Web store: http://localhost:8080/
  - Load Generator UI: http://localhost:8080/loadgen/
  - Flagd UI (feature flags): http://localhost:8080/feature/
  - Grafana: http://localhost:8080/grafana/
  - Jaeger UI: http://localhost:8080/jaeger/ui/
  - Prometheus UI: http://localhost:9090/

Section 4: OpenTelemetry Instrumentation

- Cloud Native Computing Foundation Landscape
- OpenTelemetry Instrumentation Concepts
- OpenTelemetry Language Support Matrix
- Zero-code Instrumentation Overview
- Java Zero-code Instrumentation
  - OpenTelemetry Java Instrumentation — GitHub Releases
- Python Zero-code Instrumentation
- OpenTelemetry eBPF Instrumentation (OBI) — Docs
- OBI as an OpenTelemetry Collector Receiver
- opentelemetry-ebpf-instrumentation — GitHub Repo

Examples

- OpenTelemetry Java Examples — GitHub Repo
- OpenTelemetry Python Examples — GitHub Repo
- OpenTelemetry Demo — Ad Service (Java)
- OpenTelemetry Demo — Recommendation Service (Python)

Section 5: Signals

- OpenTelemetry Signals Overview

Metrics

- Metrics Concepts
- Metrics Semantic Conventions
- Metrics SDK Specification

Traces

- Traces Concepts
- Traces Semantic Conventions
- Trace SDK Specification
- W3C Trace Context Specification

- Context Propagation Specification

Logs

- Logs Concepts
- Logs Semantic Conventions
- Logs Data Model
- Logs SDK Specification

Baggage

- Baggage Concepts
- W3C Baggage Specification

Profiling (Public Alpha)

- Profiles Concepts
- Profiles Specification
- OpenTelemetry Profiles Enters Public Alpha — Blog Post
- opentelemetry-ebpf-profiler — GitHub Repo

Tools

- Prometheus
- Grafana
- Jaeger
- OpenSearch

Section 6: SDK Architecture & Composability

- Configure the SDK — Java
- OpenTelemetry SDK Configuration — General
- Declarative Configuration
- Configuration Data Model Specification
- opentelemetry-configuration — GitHub Repo
- OpenTelemetry Specification Overview

Section 7: Standards

OTLP

- OpenTelemetry Protocol Specification
- OTLP/HTTP Specification
- OpenTelemetry Vendors

Semantic Conventions

- Semantic Conventions — Documentation
- Semantic Conventions Specification
- Semantic Conventions — GitHub Repo (source of truth)
- HTTP Semantic Conventions
- Database Semantic Conventions

Section 8: Collector

- OpenTelemetry Collector — Overview
- Collector Architecture
- Collector Configuration
- Collector Releases (binaries & installers)
- Collector Core — GitHub Repo
- Collector Contrib — GitHub Repo
- Collector Components Registry

Deployment Patterns

- Collector Deployment Patterns — Overview
- Agent Deployment Pattern
- Gateway Deployment Pattern
- Scaling the Collector
- OpenTelemetry on Kubernetes
- OpenTelemetry Operator for Kubernetes
- Horizontal Pod Autoscaling with the Collector

Extensions

- zPages Extension
  - http://localhost:55679/debug/tracez
  - http://localhost:55679/debug/servicez
  - http://localhost:55679/debug/pipelinez
  - http://localhost:55679/debug/extensionz
  - http://localhost:55679/debug/featurez

SDK Exporters

- OpenTelemetry Java SDK Exporters — GitHub Repo

Section 9: Debugging & Maintenance

- Debug Exporter
- Memory Limiter Processor
- Batch Processor
- Collector Internal Telemetry
- Collector Troubleshooting Guide
- Schema Translate Processor (Contrib)
- OpenTelemetry Schemas Specification

Section 10: SRE Principles

- Google SRE Book (free online)
- Google SRE Workbook (free online)
- Resource Semantic Conventions

Section 11: What's Next

- OpenTelemetry Documentation
- OpenTelemetry Vendors
- OpenTelemetry Registry
- OpenTelemetry Blog
- OpenTelemetry Community
- OpenTelemetry Certified Associate (OTCA) — Linux Foundation
- Awesome OpenTelemetry — GitHub Repo

Open-source Observability Backends

- Prometheus
- Grafana
- Jaeger
- OpenSearch